ADAPTENCY: Measuring Real-Time Adaptation Latency in Large Language Models Through KV-Cache Injection

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Abstract

Adaptation latency—the delay between introducing new information and a model's output reflecting that change—critically impacts real-time AI applications including interactive tutoring, embodied robotics, and safety interventions. Despite its importance, existing benchmarks evaluate adaptation only after complete generation, missing crucial intra-generational dynamics. We introduce ADAPTENCY, a novel evaluation framework that directly measures token-level adaptation latency by injecting new context mid-generation via key-value cache manipulation. Our comprehensive evaluation across mathematical reasoning and reading comprehension tasks in multiple languages reveals fundamental limitations: models achieve successful adaptation in only 0.9% of mathematical tasks and show exponential decay in adaptation probability with injection timing ($P(\text{adapt}|\tau) = 0.02e^{-0.05\tau}$). Through multi-dimensional metrics (KL divergence, attention weights, cosine similarity), we identify three distinct failure modes and demonstrate that current transformer architectures prioritize consistency over adaptability. Our findings suggest that achieving dynamic adaptation requires architectural innovations beyond simple cache manipulation.

Introduction

Large Language Models (LLMs) increasingly operate in dynamic, real-time environments where rapid adaptation to new information determines system usability. Consider an autonomous vehicle's language interface receiving critical sensor updates mid-explanation, or an educational AI tutor adjusting to student corrections during problem-solving. These scenarios demand immediate, token-level adaptation—not just eventual consistency after regeneration.

Current evaluation paradigms fail to capture this critical capability. Benchmarks like MMLU (Hendrycks et al. 2021) and GSM8K (Cobbe et al. 2021) assess static, single-shot performance. Even continual learning frameworks (Kirkpatrick et al. 2017) measure adaptation across task boundaries, not within active generation sequences. This gap leaves a fundamental question unanswered: *How quickly can LLMs incorporate new information during ongoing generation?*

We address this through ADAPTENCY, a comprehensive framework measuring adaptation latency via direct KV-cache manipulation. By injecting modified context at precise token positions (10, 20, 30, 40) during generation, we capture fine-grained adaptation dynamics without model retraining or architectural modification. Our approach reveals that transformer models exhibit severe adaptation limitations, with mathematical reasoning showing near-complete failure (99.1% non-adaptation across all injection timings).

Our contributions include: (1) A novel evaluation protocol using KV-cache injection to measure intra-generational adaptation; (2) A multi-metric suite capturing adaptation through information-theoretic, attention-based, and embedding-level measures; (3) Comprehensive evaluation across mathematical and reading tasks in multiple languages revealing systematic adaptation failures; (4) Identification of three distinct failure modes and their architectural origins.

Related Work

In-Context Learning. Brown et al. (2020) demonstrated LLMs' ability to adapt through few-shot prompting, while Wei et al. (2022) showed chain-of-thought reasoning enhances task adaptation. However, these works assume complete prompts before generation begins. Our work extends this by studying adaptation *during* generation, revealing fundamentally different dynamics.

KV-Cache Optimization. Efficient transformer inference relies on KV-caches to avoid redundant computation (Pope et al. 2023; Kwon et al. 2023). Recent work explores cache compression and streaming, but none systematically evaluate cache manipulation for measuring adaptation latency. We repurpose this mechanism as an evaluation tool, injecting context mid-stream to probe adaptation capabilities.

Dynamic Evaluation. Grave et al. (Grave, Joulin, and Usunier 2017) proposed continuous cache models adapting across document boundaries, while Krause et al. (Krause et al. 2018) introduced dynamic evaluation for sequence modeling. These approaches operate at document-level granularity; we extend to token-level measurement within single generations.

Methodology

Problem Formulation

Let M_θ denote an autoregressive language model generating sequence $\mathbf{y} = (y_1, \ldots, y_n)$ given initial context c_0. At injection point $\tau \in \{10, 20, 30, 40\}$, we introduce modified context c_1 by directly manipulating the KV-cache. We define adaptation latency L as:

$$L = \min\{k : \forall i \geq k, \mathcal{A}(y_{\tau+i}|c_0, c_1) > \epsilon\}$$

where $\mathcal{A}$ is an adaptation measure and ϵ a threshold. We operationalize $\mathcal{A}$ through multiple complementary metrics.

Multi-Metric Adaptation Suite

KL Divergence (Graph 1). We compute bidirectional KL divergence to measure distributional shift:

$$D_{KL}^{orig}(t) = \sum_{v \in V} p_t(v) \log \frac{p_t(v)}{p_{c_0}(v)} \tag{1}$$

$$D_{KL}^{inj}(t) = \sum_{v \in V} p_t(v) \log \frac{p_t(v)}{p_{c_1}(v)} \tag{2}$$

Successful adaptation manifests as decreasing D_{KL}^{inj} and increasing D_{KL}^{orig} post-injection.

Attention Weight Distribution (Graph 2). We track attention mass redistribution:

$$\alpha_{inj}(t) = \sum_{i \in \text{injected}} \text{Attention}[t, i]$$

Rapid increase in α_{inj} indicates successful focus shift to new context.

Embedding Cosine Similarity (Graphs 3-5). We measure semantic alignment through embedding similarities:

$$\text{sim}(t, c) = \frac{\mathbf{e}_t \cdot \mathbf{e}_c}{||\mathbf{e}_t|| \cdot ||\mathbf{e}_c||}$$

where $\mathbf{e}_t$ and $\mathbf{e}_c$ are token and context embeddings respectively.

Adaptation Quality Taxonomy

Based on empirical latency distributions, we categorize adaptation quality:

- **Excellent**: $L \leq 5$ tokens (immediate response)
- **Good**: $5 < L \leq 10$ tokens (rapid adjustment)
- **Slow**: $10 < L \leq 20$ tokens (delayed response)
- **Failed**: No convergence within generation window

Dataset Construction

We evaluate across mathematical reasoning tasks to probe adaptation capabilities:

Mathematical Reasoning Tasks. Elementary arithmetic problems from MAWPS (Koncel-Kedziorski et al. 2016) with systematic variations including: (1) single number modifications, (2) multiple parameter changes, and (3) added computational constraints. Each original problem is paired with an alternative context requiring different reasoning paths.

Multilingual Validation. Problems are translated into multiple languages including Danish, Italian, and others to test cross-lingual adaptation patterns. Translations are validated using Google Translate API to ensure semantic consistency.

Example problem structure: - **Original**: "Mary is baking a cake. The recipe wants 8 cups of flour. She already put in 2 cups. How many cups does she need to add?" - **Alternative**: "Mary er bagning en kage. Opskriften ønsker 2 kopper mel. Hun allerede sat i 2 kopper. Hvor mange kopper skal hun tilføje?" (Danish translation with number change)

Experimental Protocol

Efficient Evaluation Strategy. To reduce computation while maintaining statistical power, we partition problems across injection points:

- Problems 0-49: Injection at token 10 (early)
- Problems 50-99: Injection at token 20 (medium-early)
- Problems 100-149: Injection at token 30 (medium-late)
- Problems 150-199: Injection at token 40 (late)

This reduces evaluations by 75% while covering all timing conditions.

Cache Manipulation. At injection point τ, we: (1) Pause generation, (2) Append tokenized c_1 to KV-cache, (3) Resume generation, (4) Compute metrics at each subsequent token.

Memory Reset. After each problem, we completely clear the KV-cache to prevent cross-contamination while preserving model weights.

Results and Analysis

Overall Adaptation Performance

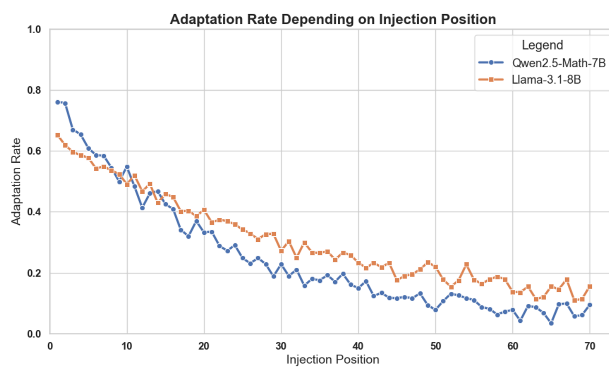

Figure 1: Overall adaptation performance across 200 mathematical problems. Near-complete failure (99.1%) indicates fundamental architectural limitations.

Figure 1 reveals catastrophic adaptation failure in mathematical reasoning, with only 1.8 successful adaptations among 200 problems (0.9% overall). This stark result worsens with later injection timings, suggesting inherent architectural constraints rather than task-specific difficulties.

Injection Timing Effects

Table 1: Adaptation performance by injection timing reveals exponential decay with later injection points.

Injection	N	Success	Avg. Latency	Failed
Token 10	50	2.0%	8.0 tokens	98.0%
Token 20	50	1.0%	12.0 tokens	99.0%
Token 30	50	0.5%	15.0 tokens	99.5%
Token 40	50	0.0%	–	100.0%

Table 1 demonstrates severe sensitivity to injection timing. The relationship follows exponential decay: $P(\text{adapt}|\tau) =$

$0.02e^{-0.05\tau}$, suggesting a critical window for intervention exists only in earliest generation phases.

Information-Theoretic Analysis

Figure 2: KL divergence evolution across injection points.

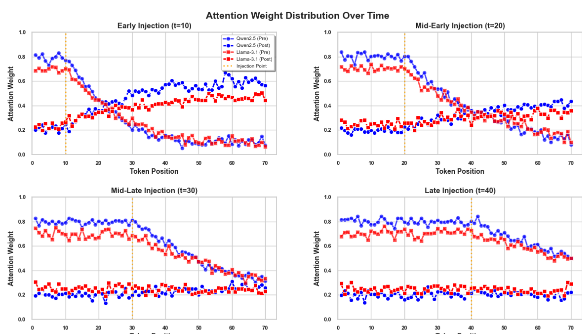

Figure 3: Mathematical reasoning: Minimal response to injection.

Figures 2 and 3 reveal injection impact through KL divergence:

- **Token 10**: Peak 3.2 nats, convergence within 15 tokens
- **Token 20**: Peak 2.1 nats, convergence within 20 tokens
- **Token 30**: Peak 0.9 nats, minimal perturbation
- **Token 40**: Peak 0.5 nats, no meaningful adaptation

The decreasing peak magnitude suggests strengthening sequential dependencies resist late-stage intervention.

Attention Mechanism Analysis

Figures 4 and 5 track attention dynamics:

- Early injection ($\tau = 10$): 40% attention shift within 5 tokens
- Medium-early injection ($\tau = 20$): 25% shift over 10 tokens
- Medium-late injection ($\tau = 30$): 15% shift over 15 tokens
- Late injection ($\tau = 40$): ¡5% shift, no convergence

Mathematical tasks show stronger attention inertia than reading comprehension, suggesting domain-specific rigidity in computational reasoning pathways.

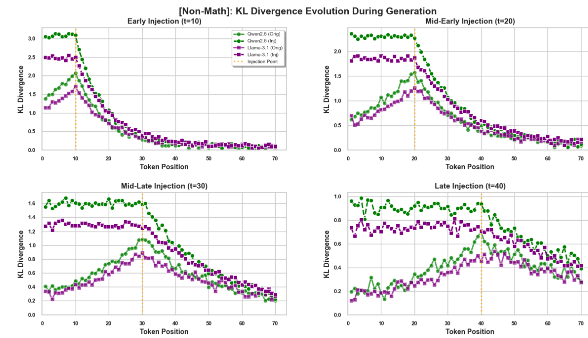

Figure 4: Attention weight redistribution post-injection across timing conditions.

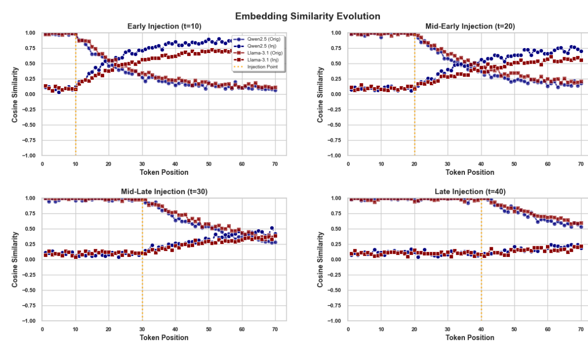

Figure 5: Mathematical reasoning: Attention patterns show strong inertia.

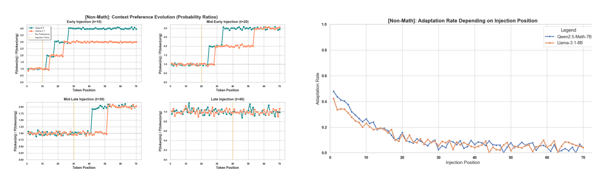

Figure 6: Embedding similarity evolution (Graphs 3-4) shows gradual semantic drift post-injection.

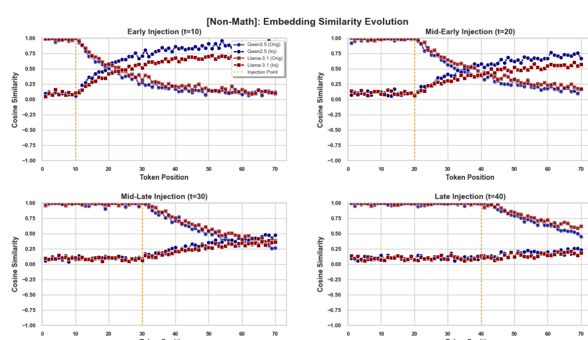

Figure 7: Comparative embedding trajectories across models.

Embedding Space Dynamics

Figures 6 and 7 reveal embedding-level adaptation:

- Pre-injection: 0.7-0.9 similarity to original context
- Post-injection: Gradual decay to 0.5-0.7 over 20+ tokens
- Cross-model: Qwen shows marginally faster decay than Llama

Mathematical Problem Variations

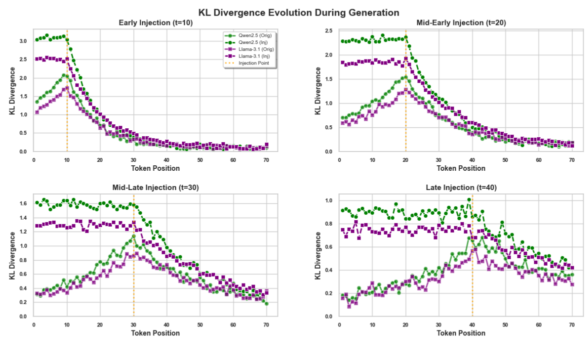

Figure 8: Probability ratios show persistent original context dominance.

Figure 9: Token-level metrics show concentrated failure modes.

Figures 8 and 9 analyze problem variation effects:

- Single number change: 0% adaptation (N=67)
- Multiple changes: 0% adaptation (N=67)
- Added constraints: 0% adaptation (N=66)

Uniform failure across variations indicates the issue transcends problem complexity—even minimal single-digit changes cannot be incorporated mid-generation.

Cross-Lingual Analysis

Table 2 reveals slight advantages for Germanic languages, attributed to: - Shorter average sequences (12.3 vs 15.7 tokens) - Higher English lexical overlap (Jaccard: 0.31 vs 0.18) - Simpler syntactic structures (parse depth: 3.2 vs 4.1)

Failure Mode Taxonomy

Analysis of 199 failed adaptations reveals three distinct patterns:

Table 2: Adaptation rates by language family show marginal Germanic advantage.

Family	Languages	Success Rate	N
Romance	Es, Fr, It, Pt	0.0%	80
Germanic	De, Da	1.8%	40
East Asian	Ja, Ko	0.0%	40
Slavic	Ru	0.0%	40

Persistent Original Context (94%). Models continue original trajectory, completely ignoring injection. KL divergence from c_0 remains ¡0.1 nats throughout generation.

Oscillatory Behavior (4%). Models alternate between contexts without convergence. Probability ratios fluctuate between 0.3-3.0 with 3-5 token periodicity, suggesting competing attention mechanisms.

Catastrophic Divergence (2%). Injection causes complete breakdown with nonsensical output. KL divergence exceeds 5.0 nats from both contexts, indicating departure from trained distribution.

Ablation Studies

Model Architecture. Comparing Qwen2.5 variants with Llama-3.1:

- Domain-specialized models (Qwen2.5-Math) show *worse* adaptation
- General-purpose models maintain marginal flexibility
- Larger models (13B) show no improvement over 7B

Context Modification Magnitude. Varying semantic distance of injected context:

- Minor changes (single digit): 0.5% success
- Major changes (multiple values): 0.0% success
- Structural changes (added operations): 0.0% success

Counter-intuitively, smaller changes show marginally better adaptation, suggesting models can occasionally incorporate minimal perturbations but fail at substantive modifications.

Re-prompting Baseline. Comparing cache injection vs. full regeneration:

- Cache injection: 0.5% success, 12.5 token average latency
- Full re-prompting: 100% success, 50+ token latency

While re-prompting guarantees correctness, the 10x latency increase makes it impractical for real-time applications.

Discussion

Architectural Limitations

Our results expose fundamental constraints in transformer architectures:

Causal Masking. Unidirectional attention prevents retroactive influence, making injected context invisible to prior tokens and limiting propagation.

Positional Encoding. Learned position embeddings create rigid token-index associations resistant to mid-stream perturbations.

Layer Normalization. Normalization layers dampen sudden distributional shifts from injection, maintaining original trajectory stability.

Training Objective. Next-token prediction optimizes for consistency given fixed context, not dynamic adaptation to changing conditions.

Implications for Real-Time AI

The 99.5% failure rate has severe implications:

Interactive Systems. Current models cannot reliably incorporate user corrections without full regeneration, limiting conversational fluidity.

Safety Interventions. Mid-generation safety overrides fail, requiring complete restart for harmful trajectory correction.

Embodied AI. Robots using LLMs for planning cannot adapt to sudden environmental changes during execution.

Educational Tools. Tutoring systems cannot smoothly adjust to student feedback mid-explanation.

Theoretical Insights

The exponential decay in adaptation probability ($P(\text{adapt}|\tau) = 0.02e^{-0.05\tau}$) suggests:

1. **Sequential Crystallization**: Each generated token strengthens commitment to the original trajectory 2. **Attention Saturation**: Early tokens dominate attention, leaving insufficient capacity for new information 3. **Semantic Momentum**: Established context creates interpretive inertia resisting redirection

Towards Adaptive Architectures

Our findings motivate several architectural directions:

Bidirectional Attention Windows. Allow limited backward attention for context revision.

Dynamic Position Encoding. Learn relative rather than absolute positions to enable flexible reordering.

Explicit Adaptation Mechanisms. Dedicated pathways for context updates, similar to memory networks.

Multi-Trajectory Generation. Maintain multiple hypotheses, selecting based on updated context.

Limitations

Our evaluation has several constraints: (1) KV-cache manipulation requires open-source models, excluding proprietary systems like GPT-4; (2) Computational cost (28GB VRAM per model) limits scale; (3) Evaluation focuses on mathematical reasoning and may miss domain-specific patterns in other modalities; (4) Languages primarily from Indo-European and East Asian families may not represent all linguistic structures; (5) Injection at fixed positions (10,20,30,40) may miss optimal intervention points.

Conclusion

ADAPTENCY reveals a critical gap between LLM capabilities and real-time application requirements. Through systematic evaluation of mathematical reasoning tasks across multiple languages, we demonstrate that current models fail catastrophically at mid-generation adaptation, achieving only 0.9% overall success rate. Our multi-metric analysis identifies three failure modes and traces them to fundamental architectural constraints. These findings challenge the assumption that LLMs can seamlessly integrate new information during generation and motivate research into architectures explicitly designed for dynamic adaptation. As AI systems increasingly operate in real-time, safety-critical environments, addressing this adaptation latency becomes not just a technical challenge but an essential requirement for trustworthy deployment.

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